Mohamed Abbas KONATE, PhD

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I am an experienced computer vision specialist and robotics engineer with a PhD in Applied Mathematics; I am passionate about innovative technologies such as Deep learning and advanced mathematical modeling. In my current role, I design intelligent AI algorithms that autonomously adapt and excel within complex, real-world changing environments in industrial robotics. I combine technical knowledge with a strong passion for research and teaching, actively mentoring future engineers. I am now seeking a new challenge in an innovative, results-oriented environment where I can contribute to the development of state-of-the-art intelligent machines and help bring pioneering solutions from concept to impactful reality.

Professional Experience

2023–Present **Expert in Computer Vision and Robotics, Michelin**, Clermont-Ferrand.

- Conducted commercial and technical feasibility analysys for internal and external clients, aligning high-TRL AI and robotics solutions with operational constraints and strategic business objectives.
- Led the end-to-end technical deployment of an autonomous tire-unloading robot from prototype to full production in client logistics facilities; Optimized perception and navigation pipelines on-site, cutting average cycle time by 37% within 4 months while boosting vision reliability from 1 error per 50 tires to 1 per 300.
- Engineered custom CUDA kernels for full-resolution point cloud processing, accelerating collision detection by 90% over previous subsampled pipelines and establishing an internal Python/CUDA library now adopted as the company-wide standard for fast collision checking.
- The impact are viable ROI, business continuity for clients while eliminating hazardous manual. Labor—saving 2 FTEs per device while ensuring strict compliance with industrial human safety standards in highly constrained environments.

2018–2023 **Researcher in Deep Learning, Robotics, and Computer Vision, Michelin**, Clermont-Ferrand.

- Designed froom scratch and implemented the company's patented first logistic autonomous tire-truck unloading robot, from initial concept to research prototype, especially the full vision system (sensor selection, perception pipeline, and integration with robotic arms such as Fanuc and Yaskawa). Up to 8 patents produced.
- Developed custom deep learning models for semantic segmentation, objects detection and classification, tailored to industrial quality control scenarios and real-time constraints (Black on black images).
- Benchmarked, selected, and integrated multi-modal sensors (2D and 3D cameras, Profilometers) with robotic platforms, enabling robust perception in black on black applications.
- Designed and integrated machine learning algorithms to automate rubber extrusion using AI in order to reduces scraps and improve cycle time. Eliminated completely the need for manual tuning of PID or RST parameters for device control using online parameter identification and predictive control.

Professional Experience (suite)

2017 – 2018 **Intern : Online Identification and Control, Michelin**, Clermont-Ferrand.
Created a predictive control system extrusion model to estimate and update parameters in real time.

Teaching

2025 – Present **Deep Learning Instructor, Institut Polytechique Félix Houphouet Boigny, Côte d'Ivoire**, Clermont-Ferrand.
Teaching Deep learning and computer vision to graduate students at INPHB, Top engineering school in West Africa.

2024 – Present **Transformers and LLMs Instructor, POLYTECH**, Clermont-Ferrand.
Teaching advanced Transformer architectures and their applications in Deep Learning (GPTs, Vision Transformerd) to graduate students.
Training engineering students in both theoretical and practical aspects of deep learning development.

Education

2021–2024 **PhD in Applied Mathematics, University Clermont Auvergne, France**
Completed doctoral research concurrently with a full-time engineering role, balancing advanced theoretical work with industrial responsibilities.
Title : *Continual Learning in Pneumatic Industry : Conditions and Applications*

- Developed both theoretical and practical advances in Continual Learning (CL) for industrial AI, bridging academic research and real-world deployment at Michelin.
- Proposed a unified theoretical framework for CL, deriving rigorous results on function approximation in high-dimensional spaces and highlighting limitations of common assumptions even with replay strategies.
- Introduced a new class of CL methods based on low-dimensional parameter-space projection for systematic function approximation, opening new research directions for scalable and robust CL algorithms.
- Designed practical CL solutions to reduce the cost and complexity of neural network lifecycle management, enabling efficient adaptation to new data and tasks without full retraining.
- Created Mean Incremental Class Accuracy (MICA), a set of metrics tailored to industrial CL evaluation, and released CLMixer, a modular open-source library for integrating and optimizing CL methods (`https://github.com/mohakonaabbas/clmixer/tree/master`).

2014 – 2017 **Double Degree : Mechanical Engineer - Master in Advanced Systems and Robotics, Arts et Métiers ParisTech - UPMC - Mines ParisTech**
Specialization : *Robotics and Computer Vision*. The curriculum emphasizes physics, materials science, solid and fluid mechanics, and advanced mathematics (analysis, optimization, numerical methods), providing a rigorous foundation that complements my major in robotics and computer vision.

2012 – 2012 **Baccalaureate C, Lycée Scientifique de Yamoussoukro, Côte d'Ivoire**
Valedictorian

Skills

Expertise Python, Deep Learning (CNNs, Transformers), Robotics, 3D Computer Vision, Algorithm Design and Optimization, PyTorch, Scientific Computing, Sensor Fusion

Skills (suite)

Intermediate	ROS2, Parallel Computing (CUDA), TensorRT, Docker, CI/CD, Version Control (Git), Web Development (React, Django), C++, Reinforcement Learning, Isaac Sim, Microsoft Azure
Languages	Fluent in English and French, Beginner in Spanish and Chinese
Soft Skills	Teamwork, Team Management, Ability to Work Under Pressure, Applied Research, Teaching, Creativity

Publications et Patents

Patents

1. M.-A. Konate, « System for Predicting Tyre Tread Profiles of Abstract Length Using Ai, » FR3168790 (A1), mai 2026. visité le 11 août 2026. adresse : https://worldwide.espacenet.com/publicationDetails/biblio?FT=D&date=20260529&DB=&locale=fr_EP&CC=FR&NR=3168790A1&KC=A1&ND=6.

2. O. Ludcher, F. Buisson, M.-A. Konate, T. Simonelli et J. Chagnard, « Procédé D'usinage De La Couche Externe D'un Bandage Au Cours Duquel on Adapte La Trajectoire De Coupe En Fonction De L'orientation Des Fils De Renfort Qui Sont Situés Sous Ladite Couche Externe, » FR3151780 (A1), fév. 2025. visité le 11 août 2026. adresse : https://worldwide.espacenet.com/publicationDetails/biblio?FT=D&date=20250207&DB=&locale=fr_EP&CC=FR&NR=3151780A1&KC=A1&ND=5.

3. F. Da-Costa, P. Bouges, M.-A. Konate et N. Bard, « Chaining of Tyres by Robotic Means Using Three-Dimensional Viewing, » WO2024231119 (A1), nov. 2024. visité le 11 août 2026. adresse : https://worldwide.espacenet.com/publicationDetails/biblio?FT=D&date=20241114&DB=&locale=fr_EP&CC=WO&NR=2024231119A1&KC=A1&ND=5.

4. J.-M. Dettorre, N. Bard, Q. Deniau, M. Druet et M.-A. Konate, « System and Process for Picking Tires in an Unknown Arrangement, » US2024075626 (A1), mars 2024. visité le 11 déc. 2024. adresse : https://worldwide.espacenet.com/publicationDetails/biblio?FT=D&date=20240307&DB=&locale=fr_EP&CC=US&NR=2024075626A1&KC=A1&ND=4.

5. J.-M. Dettorre, N. Bard, M.-A. Konate, P. Bouges et R. Calvel, « System and Method for Acquiring and Determining Axes of Valves in Tyre Vulcanisation Moulds, » WO2024012960 (A1), jan. 2024. visité le 11 déc. 2024. adresse : https://worldwide.espacenet.com/publicationDetails/biblio?FT=D&date=20240118&DB=&locale=fr_EP&CC=WO&NR=2024012960A1&KC=A1&ND=4.

6. J.-M. Dettorre, N. Bard, M.-A. Konate, P. Bouges et R. Calvel, « System and Method for Acquiring and Determining Valve Axes in Tyre Vulcanisation Moulds, » WO2024089157 (A1), mai 2024. visité le 11 déc. 2024. adresse : https://worldwide.espacenet.com/publicationDetails/biblio?FT=D&date=20240502&DB=&locale=fr_EP&CC=WO&NR=2024089157A1&KC=A1&ND=4.

7. J.-M. Dettorre, M.-A. Konate, N. Bard, K. Bouvet et F. Da-Costa, « System and Method for Managing the Loading and Unloading of Tyres in an Unknown Arrangement, » WO2024125875 (A1), juin 2024. visité le 11 déc. 2024. adresse : https://worldwide.espacenet.com/publicationDetails/biblio?FT=D&date=20240620&DB=&locale=fr_EP&CC=WO&NR=2024125875A1&KC=A1&ND=4.

8. M.-A. Konate, « Process and System for Controlling a Gripping Device Used in Cycles for Sorting Tires Arranged in Unknown Arrangements, » EP4444510 (A1), oct. 2024. visité le 11 déc. 2024. adresse : https://worldwide.espacenet.com/publicationDetails/biblio?FT=D&date=20241016&DB=&locale=fr_EP&CC=EP&NR=4444510A1&KC=A1&ND=4.

9. T. Chateau, P. Phutane et M.-A. Konate, « System and Process for Automatic Interface Recognition in Tire Product Profiles, » US2023281976 (A1), sept. 2023. visité le 11 déc. 2024. adresse : https://worldwide.espacenet.com/publicationDetails/biblio?FT=D&date=20230907&DB=&locale=fr_EP&CC=US&NR=2023281976A1&KC=A1&ND=4.

10 M. Druet, Q. Deniau, M.-A. Konate, J.-M. Dettorre et N. Bard, « Système et Procédé de Dévracage des Pneumatiques dans un Arrangement Inconnu, » FR3118003 (A1), juin 2022. visité le 11 août 2026. adresse : https://worldwide.espacenet.com/publicationDetails/biblio?FT=D&date=20220624&DB=&locale=fr_EP&CC=FR&NR=3118003A1&KC=A1&ND=5.